

PROJECT ANALYSIS

CAPITAL BUDGETING DECISIONS

PROJECT CASH FLOWS

- Capital budgeting is concerned primarily with the after-tax **net (operating) cash flows** (NCF) of a particular project, or change in cash inflows minus change in cash outflows.

$$NCF = \Delta OEAT + \Delta Dep - \Delta NWC$$

operating earnings after taxes, $\Delta OEAT$, plus the change in depreciation, ΔDep , minus the change in the net working capital investment required by the firm to support the project, ΔNWC :⁹

$$NCF = (\Delta R - \Delta O - \Delta Dep)(1 - T) + \Delta Dep - \Delta NWC$$

CAPITAL INVESTMENT: IMPORTANCE

- Long term Effects
- Irreversibility
- Substantial Outlay

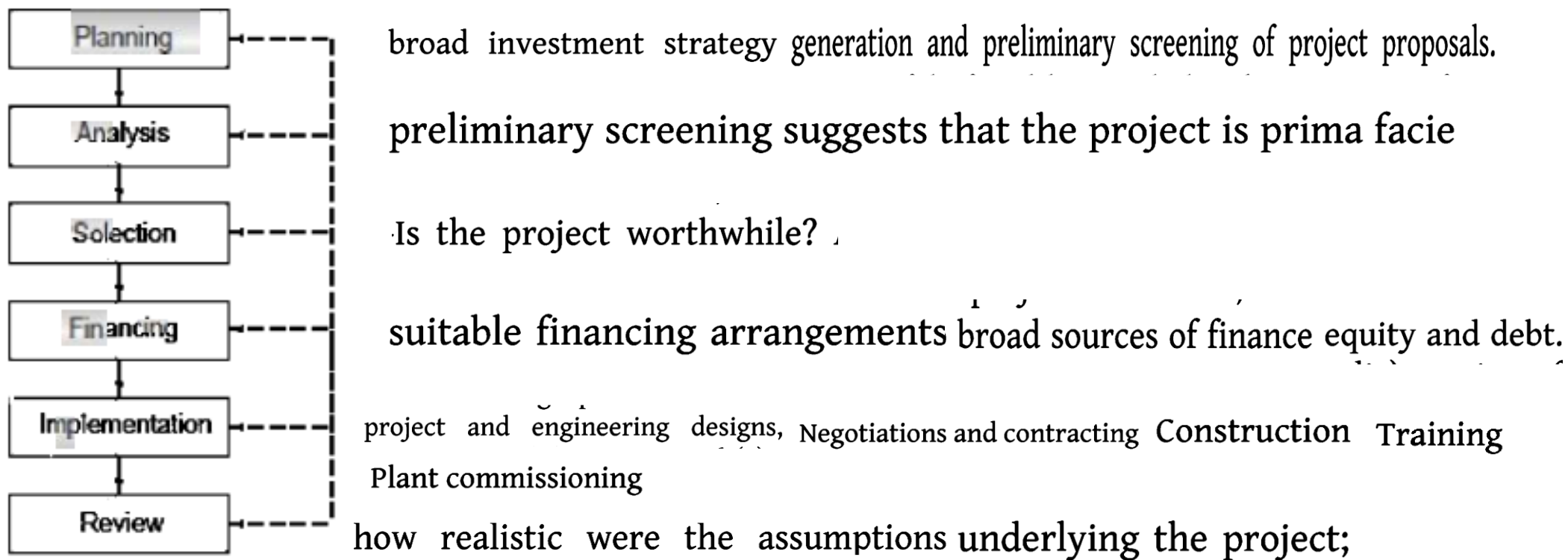
ISSUES WITH CAPITAL BUDGETING

- Measurement Problems
- Uncertainty
- Temporal spread

PROJECT CLASSIFICATION

- Mandatory Investments
- Replacement Projects
- Expansion Projects
- Diversification Projects
- Research and Development Projects
- Miscellaneous Projects

PROCESS OF CAPITAL BUDGETING



FACETS OF PROJECT ANALYSIS

The important facets of project analysis are:

- Market analysis
- Technical analysis
- Financial analysis
- Economic analysis
- Ecological analysis

aggregate demand market share

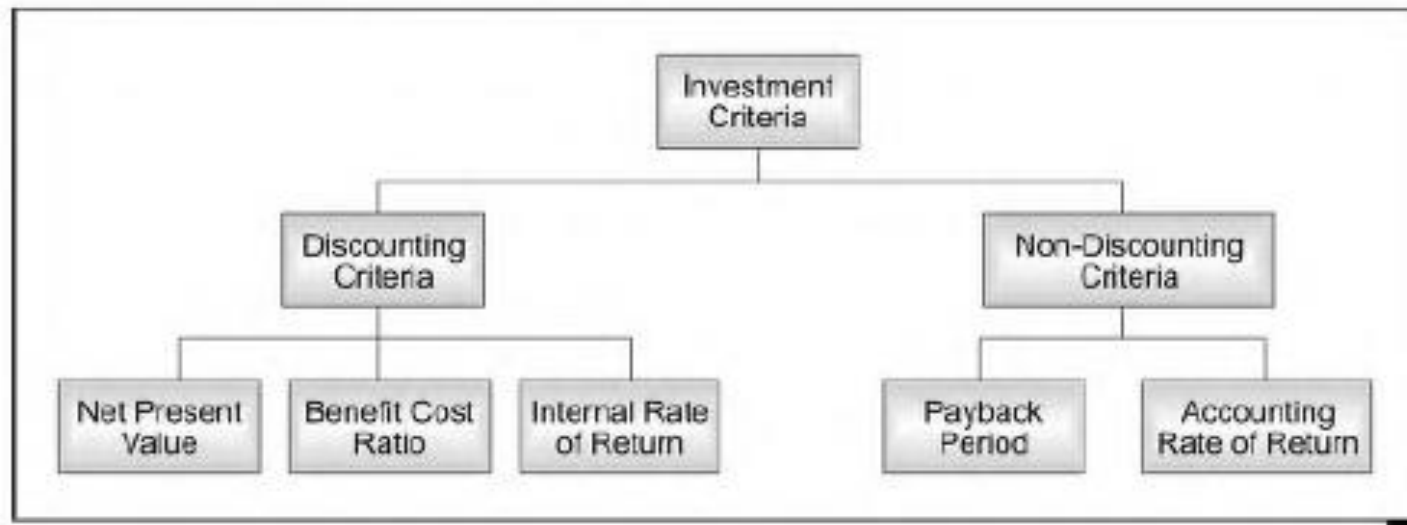
technical and engineering aspects

proposed project will be financially viable

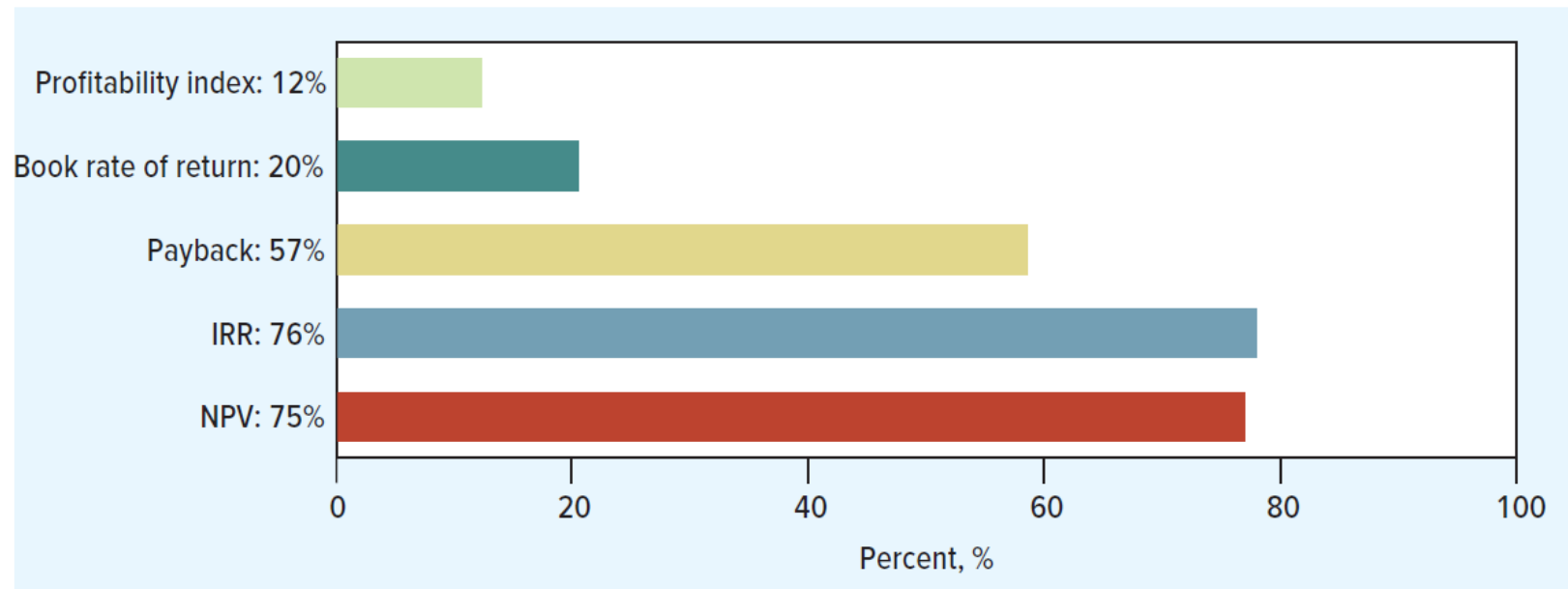
social cost benefit analysis

environmental concerns

INVESTMENT CRITERIA



USE OF TECHNIQUES



Source: J. R. Graham and C. R. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001), pp. 187–243.

NET PRESENT VALUE (NPV)

$$NPV = \sum_{t=1}^n \frac{C_t}{(1+r)^t} - \text{Initial investment}$$

C_t is the cash flow at the end of year t
 n is the life of the project, and
 r is the discount rate.

An investment should be accepted if the net present value is positive and rejected if it is negative.

Advantage of NPV

1. Net Present Values Are Additive

$$NPV(A + B) = NPV(A) + NPV(B)$$

2. Intermediate Cash Flows Are Invested at the Cost of Capital
3. NPV Calculation Permits Time Varying Discount Rates

$$NPV = \sum_{t=1}^n \frac{C_t}{\prod_{j=1}^t (1+r_j)} - \text{Initial investment}$$

Discount rate	14%	15%	16%	18%	20%
Investment	-12000				
Cash flow	4,000	5,000	7,000	6,000	5,000

EXAMPLE

<i>Year</i>	<i>Cash flow</i>
0	Rs.(1,000,000)
1	200,000
2	200,000
3	300,000
4	300,000
5	350,000

The cost of capital¹, r , for the firm is 10 percent.

SOLUTION

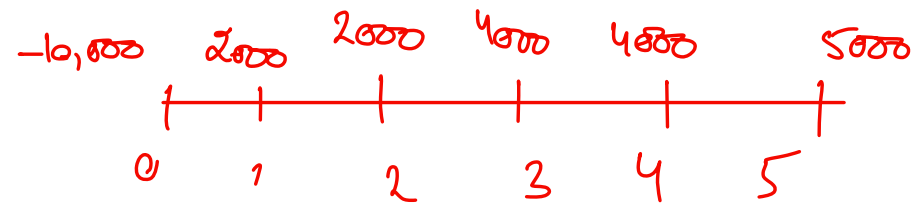
$$\begin{aligned}\text{NPV} &= -\frac{200,000}{(1.10)^1} + \frac{200,000}{(1.10)^2} + \frac{300,000}{(1.10)^3} + \frac{300,000}{(1.10)^4} + \frac{350,000}{(1.10)^5} - 1,000,000 \\ &= -5,272\end{aligned}$$

NPV- DISADVANTAGE

- The NPV is expressed in absolute terms rather than relative terms and hence does not factor in the scale of investment.

	project A	project B
NPV	₹ 5,000	₹ 2,500
Investment	₹ 50,000	₹ 10,000

- The NPV rule does not consider the life of the project



EXERCISE

$$NPV = \frac{2000}{1.1^1} + \frac{2000}{(1.1)^2} + \frac{4000}{(1.1)^3} + \frac{4000}{(1.1)^4} + \frac{5000}{(1.1)^5} = 12,312.994$$

Suppose we are asked to decide whether a new consumer product should be launched. Based on projected sales and costs, we expect that the cash flows over the five-year life of the project will be \$2,000 in the first two years, \$4,000 in the next two, and \$5,000 in the last year. It will cost about \$10,000 to begin production. We use a 10 percent discount rate to evaluate new products. What should we do here?

$$NPV = 12,313 - 10,000 = 2313 \quad \text{Yes}$$

SOLUTION

$$\begin{aligned}\text{Present value} &= (\$2,000/1.1) + (2,000/1.1^2) + (4,000/1.1^3) \\ &\quad + (4,000/1.1^4) + (5,000/1.1^5) \\ &= \$1,818 + 1,653 + 3,005 + 2,732 + 3,105 \\ &= \$12,313\end{aligned}$$

The present value of the expected cash flows is \$12,313, but the cost of getting those cash flows is only \$10,000, so the NPV is $\$12,313 - 10,000 = \$2,313$. This is positive; so, based on the net present value rule, we should take on the project.

BENEFIT COST RATIO (PROFITABILITY INDEX)

Benefit-cost ratio : $BCR = \frac{PVB}{I}$

Net benefit-cost ratio: $NBCR = \frac{PVB - I}{I} = BCR - 1$

Decision Rules

When BCR	or NBCR	Rule is
>1	>0	Accept
=1	=0	Indifferent
<1	<0	Reject

Initial investment:		Rs.100,000
Benefits:	Year 1	25,000
	Year 2	40,000
	Year 3	40,000
	Year 4	50,000

$$\begin{aligned}
 NPV &= \frac{25,000}{1.12} + \frac{40,000}{(1.12)^2} + \frac{40,000}{(1.12)^3} \\
 &\quad + \frac{50,000}{(1.12)^4} - 1,00,000 \\
 &= 14,456.29
 \end{aligned}$$

cost of capital of 12 percent.

SOLUTION

The benefit cost ratio measures for this project are:

$$\text{BCR} = \frac{\frac{25,000}{(1.12)} + \frac{40,000}{(1.12)^2} + \frac{40,000}{(1.12)^3} + \frac{50,000}{(1.12)^4}}{100,000} = 1.145$$

$$\text{NBCR} = \text{BCR} - 1 = 0.145$$

INTERNAL RATE OF RETURN (IRR)

- The internal rate of return (IRR) of a project is the discount rate which makes its NPV equal to zero.
- *The discount rate that makes $NPV = 0$ is also the rate of return.*

$$NPV = \sum_{t=1}^n \frac{C_t}{(1+r)^t} - \text{Initial investment}$$

$$NPV = -C_0 + \frac{C_1}{1 + IRR} + \frac{C_2}{(1 + IRR)^2} + \dots + \frac{C_T}{(1 + IRR)^T} = 0$$

EXAMPLE

Cash Flows (\$)		
C_0	C_1	C_2
-4,000	+2,000	+4,000

$$NPV = -4,000 + \frac{2,000}{1 + IRR} + \frac{4,000}{(1 + IRR)^2} = 0$$

$$NPV = -4,000 + \frac{2,000}{1.0} + \frac{4,000}{(1.0)^2} = +\$2,000$$

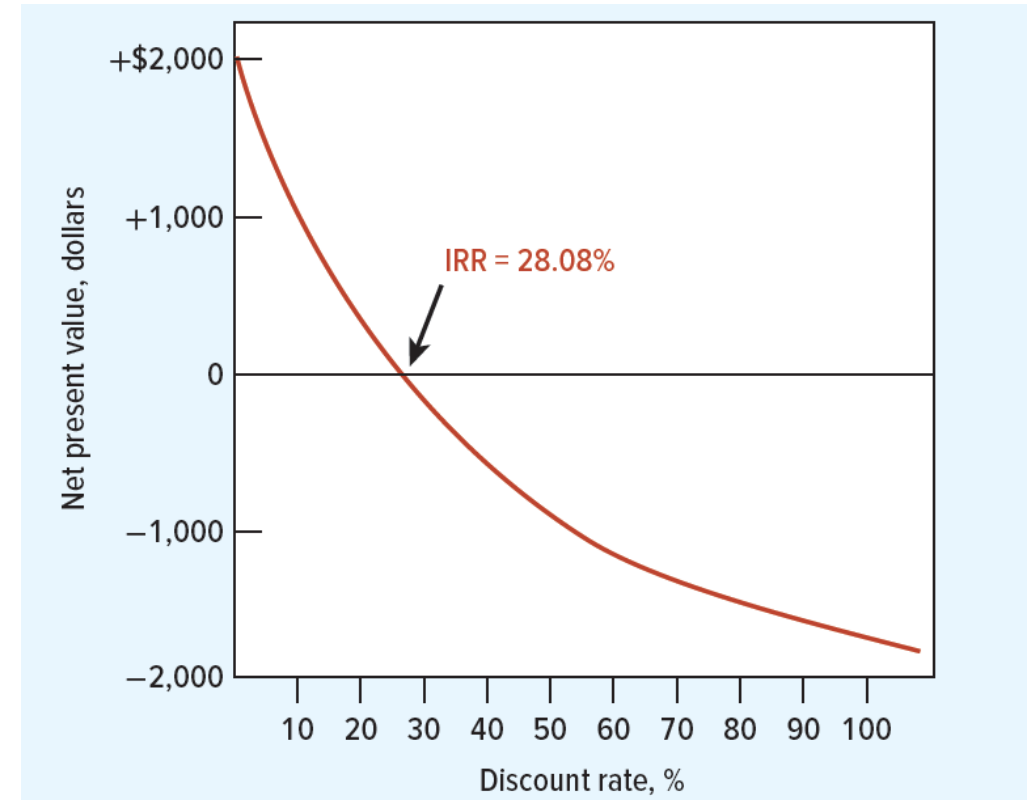
$$NPV = -4,000 + \frac{2,000}{1.50} + \frac{4,000}{(1.50)^2} = -\$889$$

therefore, the IRR must be less than 50%.

try a zero discount rate.

try a discount rate of 50%.

IRR is 28.08%.



Acceptance Rule

The firm should accept an investment project if the opportunity cost of capital is less than the internal rate of return

EXERCISE

Consider the cash flows of a project being evaluated by Techtron Limited:

<i>Year</i>	0	1	2	3	4
<i>Cash flow</i>	(100,000)	30,000	30,000	40,000	45,000

Year	Cash Flow	15.000%	12.000%	15.365%
0	1,00,000			
1	30,000	26086.957	26785.71	26004.421
2	30,000	22684.31	23915.82	22540.997
3	40,000	26300.649	28471.21	26051.803
4	45,000	25728.896	28598.31	25404.826
	NPV	801	7,771	2

INTERPOLATION

Interpolation	IRR	NPV
1	15.000%	801
2	x	0
3	15.500%	-291

Using (2-1)	15- x	801- 0'
Using (3-1)	15.50' - 15.0'	<u>-291 - 801</u>

$\frac{0.15 - x}{0.005} = \frac{801}{-1,092}$

0.15-x =	-0.00366758
x=	0.153667582

15.3668%

ISSUES WITH IRR?

- Not all cash-flow streams have NPVs that decline as the discount rate increases.

Cash Flows (\$)				
Project	C_0	C_1	IRR	NPV at 10%
A	-1,000	+1,500	+50%	+364
B	+1,000	-1,500	+50%	-364

ISSUES WITH IRR?

- **Multiple Rates of Return**
- A mining firm involves an initial investment of A\$30 billion and is expected to produce a cash inflow of A\$10 billion a year for the next nine years. At the end of that time, the company will incur A\$65 billion of cleanup costs.

Cash Flows (billions of Australian dollars)				
C_0	C_1	...	C_9	C_{10}
-30	10		10	-65

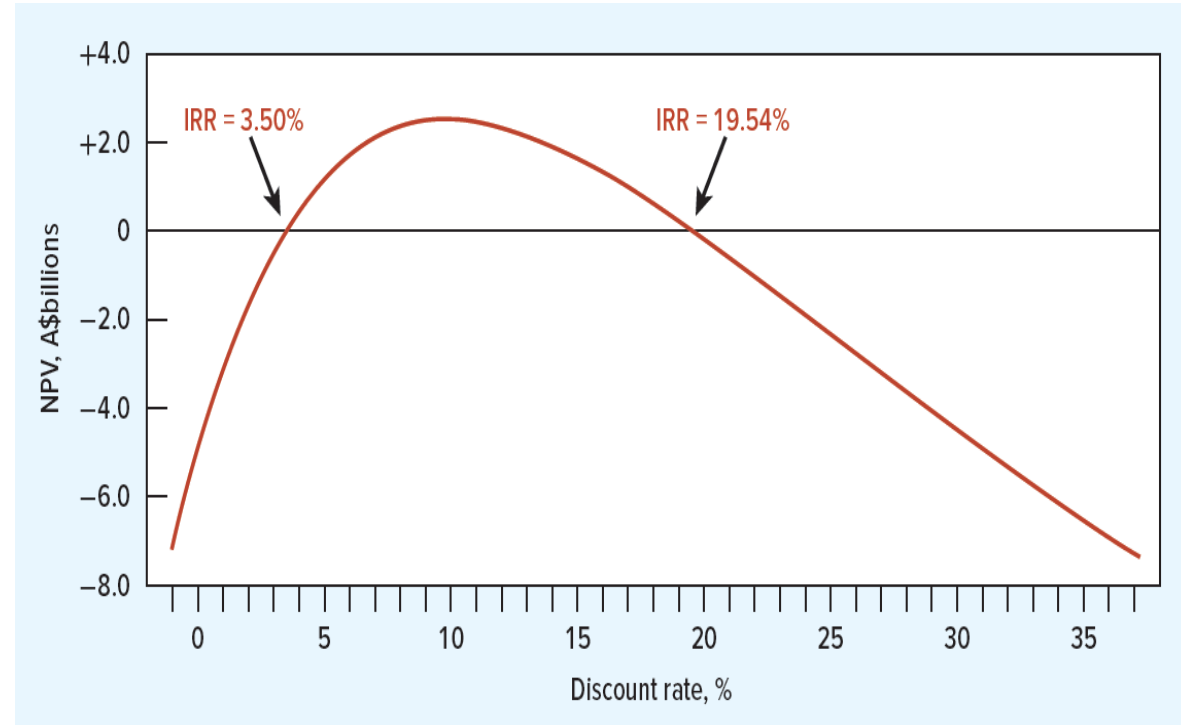
IRR (%)	NPV at 10%
+3.50 and 19.54	\$A2.53 billion

MULTIPLE IRR

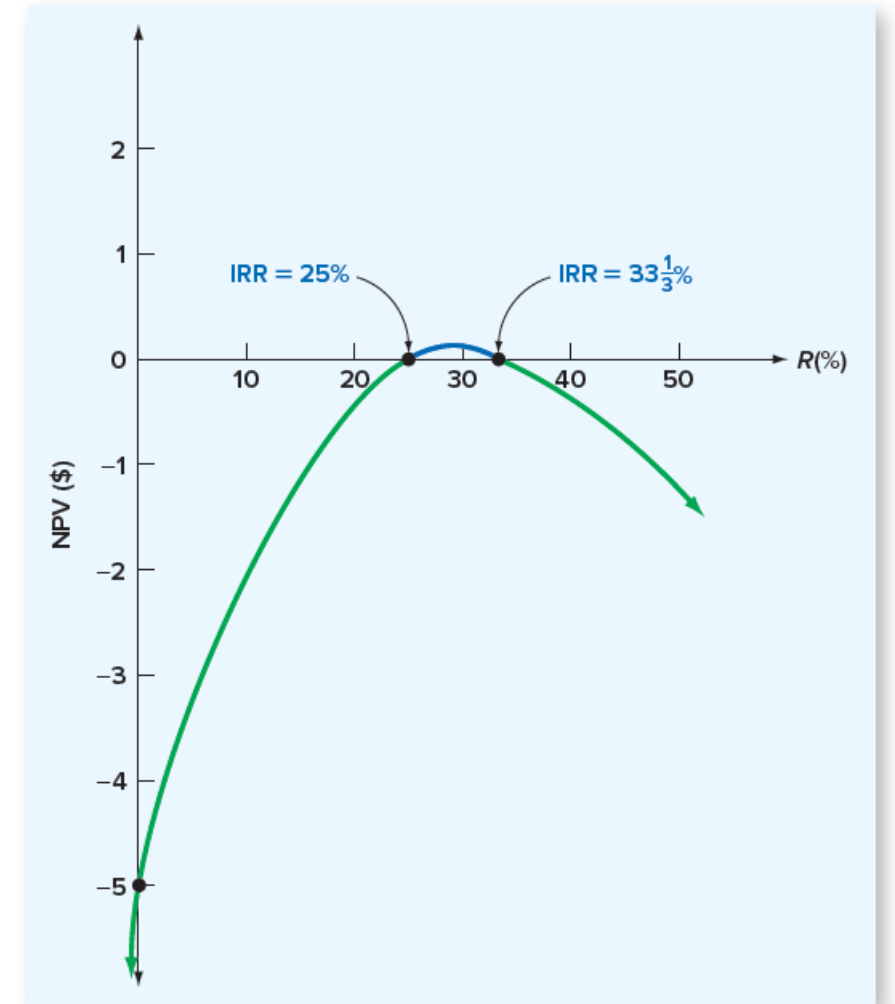
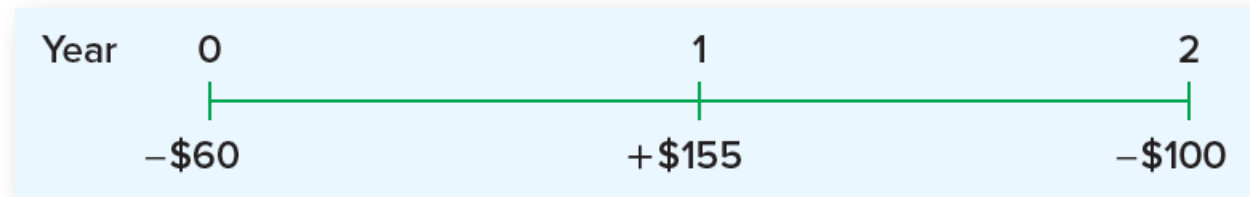
$$NPV = -30 + \frac{10}{1.035} + \frac{10}{1.035^2} + \dots + \frac{10}{1.035^9} - \frac{65}{1.035^{10}} = 0$$

$$NPV = -30 + \frac{10}{1.1954} + \frac{10}{1.1954^2} + \dots + \frac{10}{1.1954^9} - \frac{65}{1.1954^{10}} = 0$$

IRR of both 3.50% *and* 19.54%.



ANOTHER EXAMPLE

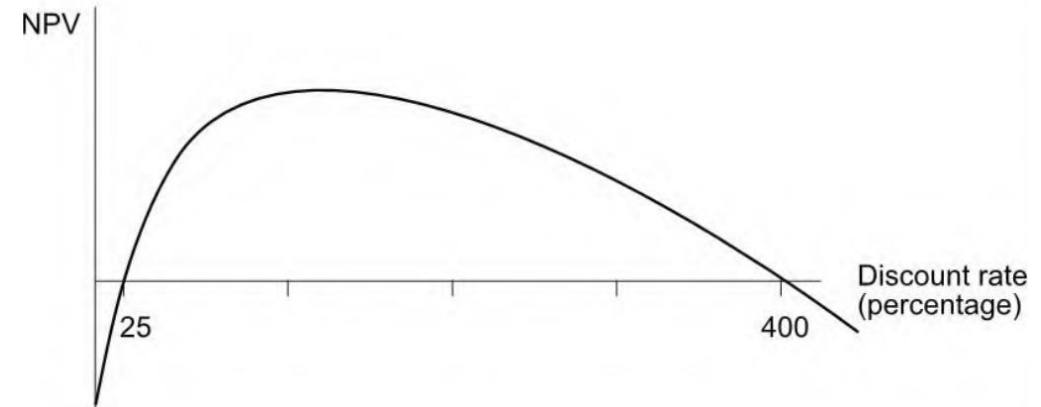


ANOTHER EXAMPLE

Non-conventional Cash Flows

Project	Cash flow		
	C_0	C_1	C_2
M	-160,000	+1,000,000	-1,000,000

$$-160,000 + \frac{1,000,000}{(1+r)} - \frac{1,000,000}{(1+r)^2} = 0$$



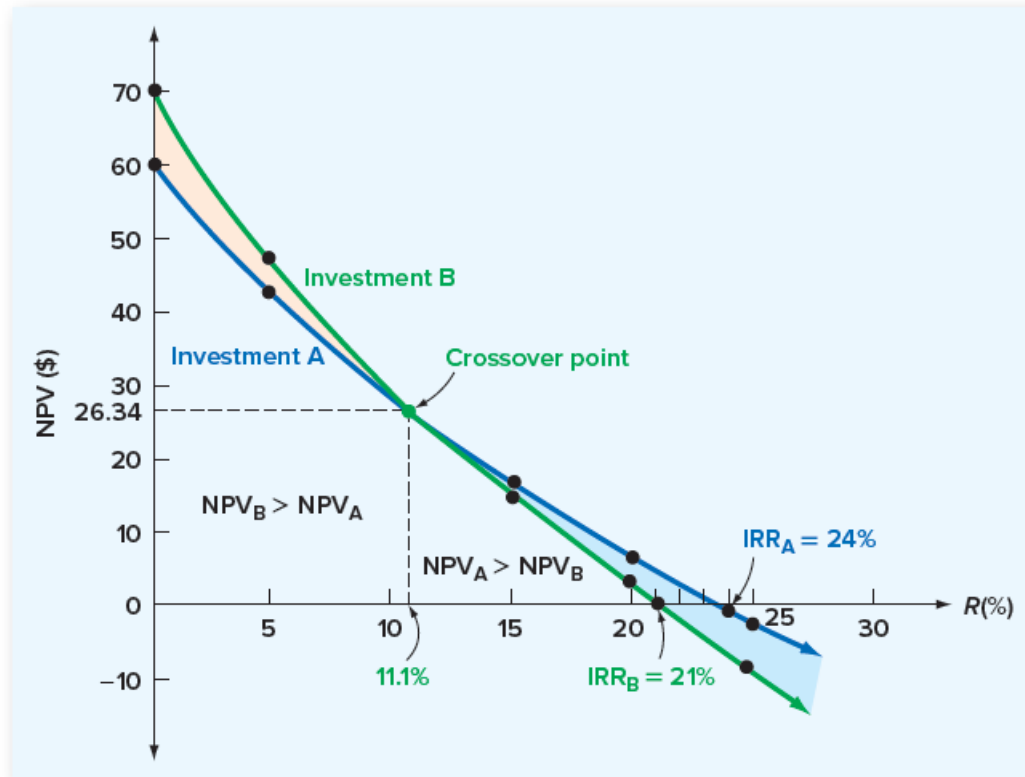
MUTUALLY EXCLUSIVE PROJECTS

A situation in which taking one investment prevents the taking of another.

Year	Investment A	Investment B
0	-\$100	-\$100
1	50	20
2	40	40
3	40	50
4	30	60

Discount Rate	NPV(A)	NPV(B)
0%	\$60.00	\$70.00
5	43.13	47.88
10	29.06	29.79
15	17.18	14.82
20	7.06	2.31
25	- 1.63	- 8.22

NPV PROFILES FOR MUTUALLY EXCLUSIVE INVESTMENTS



ANOTHER EXAMPLE

IRR is also unreliable for ranking projects which have different patterns of cash flow over time.

<i>Project</i>	C_0	C_1	C_2	C_3	C_4	<i>IRR</i>	<i>NPV at 10%</i>
X	-110,000	+31,000	+40,000	+50,000	+70,000	22%	36,613
Y	-110,000	+71,000	+40,000	+40,000	+20,000	25%	31,314

cost of capital of
10 per cent.

METHOD UNDER CAPITAL RATIONING

opportunity cost of capital is 10%

Cash Flows (\$ millions)				
Project	C_0	C_1	C_2	NPV at 10%
A	-10	+30	+5	21
B	-5	+5	+20	16
C	-5	+5	+15	12

limited to spending \$10 million

When money is limited, we must pick the projects that offer the **highest net present value per dollar** of initial outlay. This ratio is known as the **profitability index**:

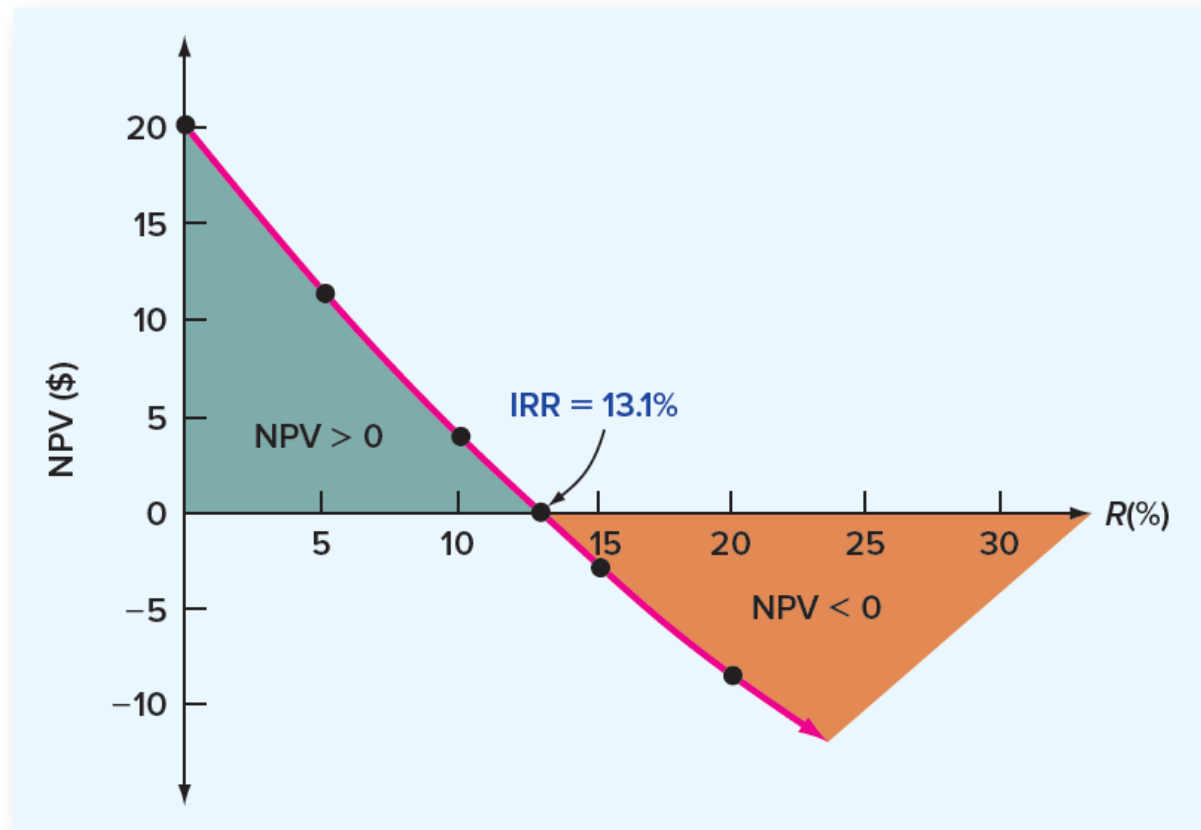
$$\text{Profitability index} = \frac{\text{net present value}}{\text{investment}}$$

Project	Investment (\$ millions)	NPV (\$ millions)	Profitability Index
A	10	21	2.1
B	5	16	3.2
C	5	12	2.4

NPV VS IRR



Discount Rate	NPV
0%	\$20.00
5	11.56
10	4.13
15	- 2.46
20	- 8.33



PROS AND CONS: IRR

Pros

- Closely related to NPV
- Easy to understand and interpret

Cons

- May lead to multiple rates of return
- May result into incorrect decisions in comparing mutually exclusive projects

MIRR: MODIFIED INTERNAL RATE OF RETURN

The procedure for calculating MIRR is as follows:

Step 1: Calculate the present value of the costs (PVC) associated with the project, using the cost of capital (r) as the discount rate:

$$\text{PVC} = \sum_{t=0}^n \frac{\text{Cash outflow}_t}{(1+r)^t} \quad (11.7)$$

Step 2: Calculate the terminal value (TV) of the cash inflows expected from the project:

$$\text{TV} = \sum_{t=0}^n \text{Cash inflow}_t (1+r)^{n-t} \quad (11.8)$$

Step 3: Obtain MIRR by solving the following equation:

$$\text{PVC} = \frac{\text{TV}}{(1+\text{MIRR})^n} \quad (11.9)$$

EXAMPLE

<i>Year</i>	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>
Cash flow	-120	-80	20	60	80	100	120

cost of capital is 15 percent.

The present value of costs is:

$$120 + \frac{80}{(1.15)} = 189.6$$

The MIRR is obtained as follows:

$$189.6 = \frac{467}{(1 + \text{MIRR})^6}$$

$$(1 + \text{MIRR})^6 = 2.463$$

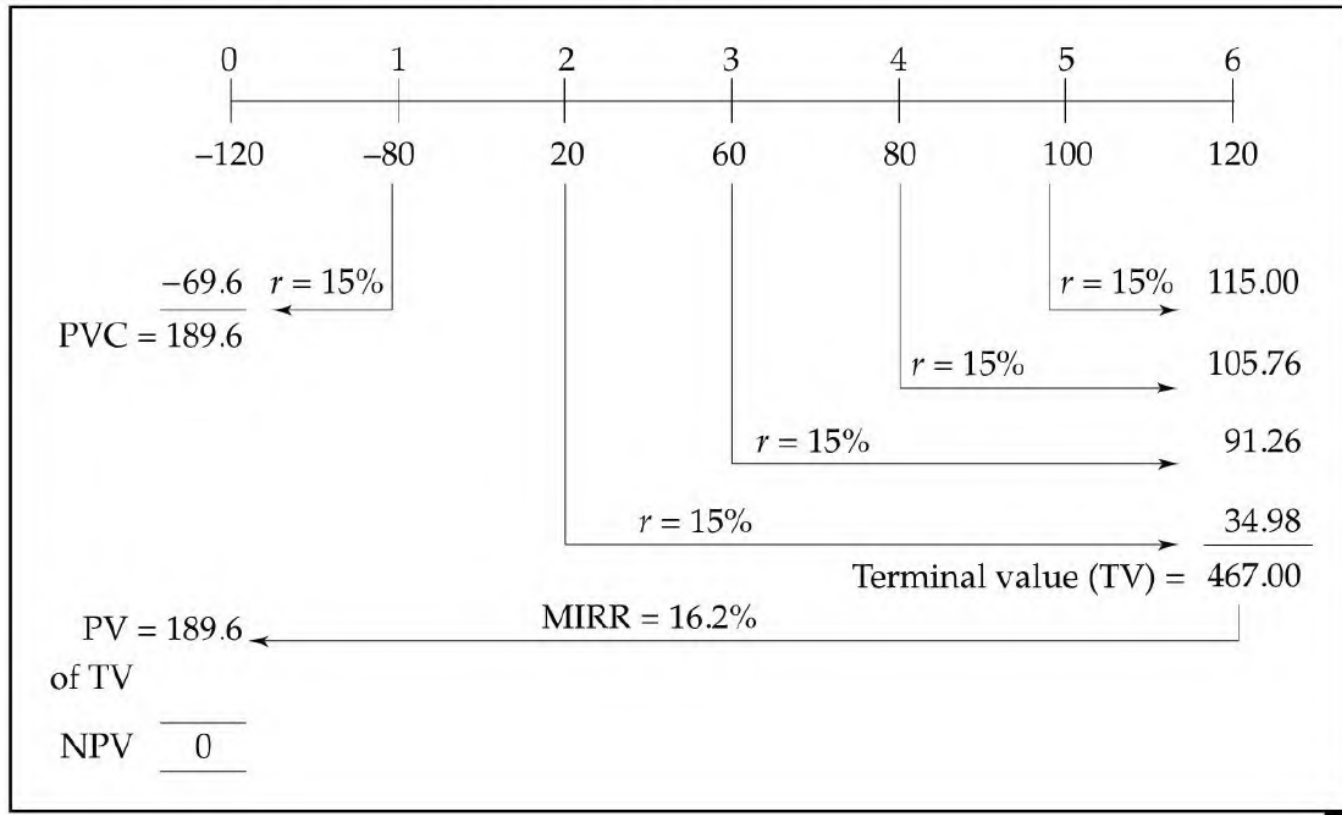
$$1 + \text{MIRR} = 2.463^{1/6} = 1.162$$

$$\text{MIRR} = 1.162 - 1 = 0.162 \text{ or } 16.2 \text{ percent}$$

The terminal value of cash inflows is:

$$20(1.15)^4 + 60(1.15)^3 + 80(1.15)^2 + 100(1.15) + 120 \\ = 34.98 + 91.26 + 105.76 + 115 + 120 = 467$$

FLOW DIAGRAM



BOOK (ACCOUNTING) RATE OF RETURN

- Ratio of book income from the investment and the book value of the assets that the firm is proposing to acquire:

$$\text{Book rate of return} = \frac{\text{book income}}{\text{book assets}}$$

$$\text{Average Accounting Return} = \frac{\text{Profit after tax}}{\text{Book value of the investment}} = \frac{\text{Average net income}}{\text{Average book value}}$$

Based on the average accounting return rule, a project is acceptable if its average accounting return exceeds a target average accounting return.

EXAMPLE

<i>Year</i>	<i>Book value of investment</i>	<i>Profit after tax</i>
1	Rs.90,000	Rs.20,000
2	80,000	22,000
3	70,000	24,000
4	60,000	26,000
5	50,000	28,000

accounting rate of return is: $\frac{1/5(20,000 + 22,000 + 24,000 + 26,000 + 28,000)}{1/5(90,000 + 80,000 + 70,000 + 60,000 + 50,000)} = 34 \text{ percent}$

accounting rate of return equal to or greater than a pre-specified cut-off

SHORTCOMINGS

- It is based upon accounting profit, not cash flow.
- It does not take into account the time value of money.
- The accounting rate of return measure is internally inconsistent.
- The accounting rate of return does not provide any guidance on what the target rate of return should be.

<i>A</i>					<i>B</i>			
<i>Year</i>	<i>Book value</i>	<i>Depreciation</i>	<i>Profit after tax</i>	<i>Cash flow</i>	<i>Book value</i>	<i>Depreciation</i>	<i>Profit after tax</i>	<i>Cash flow</i>
0	100,000	0	0	(100,000)	100,000	0	0	(100,000)
1	75,000	25,000	40,000	65,000	75,000	25,000	10,000	35,000
2	50,000	25,000	30,000	55,000	50,000	25,000	20,000	45,000
3	25,000	25,000	20,000	45,000	25,000	25,000	30,000	55,000
4	0	25,000	10,000	35,000	0	25,000	40,000	65,000

PAYBACK PERIOD

- The payback period is the length of time required to recover the initial cash outlay on the project.
- For example, if a project involves a cash outlay of Rs.600,000 and generates cash inflows of Rs.100,000, Rs.150,000, Rs.150,000, and Rs.200,000, in the first, second, third, and fourth years, respectively, its payback period is 4 years because the sum of cash inflows during 4 years is equal to the initial outlay.

Based on the payback rule, an investment is acceptable if its calculated payback period is less than some prespecified number of years.

ADVANTAGE AND DISADVANTAGE

- It is simple, both in concept and application.
 - It favours projects which generate substantial cash inflows in earlier years and discriminates against projects which bring substantial cash inflows in later years but not in earlier years.
 - Since it emphasises earlier cash inflows, it may be a sensible criterion when the firm is pressed with problems of liquidity.
- It fails to consider the time value of money. Cash inflows, in the payback calculation, are simply added without suitable discounting.
 - It ignores cash flows beyond the payback period. This leads to discrimination against projects which generate substantial cash inflows in later years.

EXAMPLE

<i>Year</i>	<i>Cash flow of A</i>	<i>Cash flow of B</i>
0	Rs. (100,000)	Rs. (100,000)
1	50,000	20,000
2	30,000	20,000
3	20,000	20,000
4	10,000	40,000
5	10,000	50,000
6	—	60,000

The payback criterion prefers A, which has a payback period of 3 years in comparison to B which has a payback period of 4 years, even though B has very substantial cash inflows in years 5 and 6.

DISCOUNTED PAYBACK PERIOD

The discounted payback period is the length of time until the sum of the discounted cash flows is equal to the initial investment.

Cash Flows (\$)						
Project	C_0	C_1	C_2	C_3	Payback Period (years)	NPV at 10%
A	-2,000	500	500	5,000	3	+2,624
B	-2,000	500	1,800	0	2	-58
C	-2,000	1,800	500	0	2	-50

Discounted Cash Flows (\$)						
Project	C_0	C_1	C_2	C_3	Discounted Payback Period (years)	NPV at 10%
A	-2,000	$500/1.10 = 455$	$500/1.10^2 = 413$	$5,000/1.10^3 = 3,757$	3	+2,624
B	-2,000	$500/1.10 = 455$	$1,800/1.10^2 = 1,488$		—	-58
C	-2,000	$1,800/1.10 = 1,636$	$500/1.10^2 = 413$		2	+50

ANOTHER EXAMPLE

<i>Year</i>	<i>Cash Flow</i>	<i>Discounting Factor @10%</i>	<i>Present Value</i>	<i>Cumulative Net Cash Flow after Discounting</i>
0	-10,000	1.000	-10,000	-10,000
1	3,000	0.909	2,727	- 7,273
2	3,000	0.826	2,478	- 4,795
3	4,000	0.751	3,004	- 1,791
4	4,000	0.683	2,732	941
5	5,000	0.621	3,105	
6	2,000	0.565	1,130	
7	3,000	0.513	1,539	

SUMMARY

	Net Present Value	Benefit Cost Ratio	Internal Rate of Return	Payback Period	Accounting Rate of Return
<i>Theoretical considerations</i>					
1. Does the method consider all cash flows	Yes	Yes	Yes	No	?
2. Does the method discount cash flows at the opportunity cost of funds?	Yes	Yes	No	No	No
3. Does the method satisfy the principle of value additivity?	Yes	No	No	?	?
4. From a set of mutually exclusive projects, does the method choose the projects which maximise shareholder wealth?	Yes	No	No	?	?
<i>Practical considerations</i>					
1. Is the method simple?	Yes	Yes	Yes	Yes	Yes
2. Can the method be used with limited information?	No	No	No	Perhaps	Yes
3. Does the method give a relative measure?	No	Yes	Yes	No	Yes

EXERCISE

The expected cash flows of a project are as follows:

<i>Year</i>	<i>Cash flow</i>
0	– 100,000
1	20,000
2	30,000
3	40,000
4	50,000
5	30,000

The cost of capital is 12 percent. Calculate the following:

- (a) net present value,
- (b) benefit-cost ratio,
- (c) internal rate of return,
- (d) modified internal rate of return,
- (e) payback period, and
- (f) discounted payback period.

NPV

a. The net present value is:

$$\begin{aligned} & -100,000 + 20,000 / (1.12) + 30,000 / (1.12)^2 + 40,000 / (1.12)^3 + 50,000 / (1.12)^4 \\ & + 30,000 / (1.12)^5 \\ & = -100,000 + 17,860 + 23,910 + 28,480 + 31,800 + 17,010 = 19,060 \end{aligned}$$

SOLUTION

b. The benefit-cost ratio is:

$$119,060 / 100,000 = 1.19$$

c. Try a discount rate of 18 percent. The NPV at 18 percent discount rate is 1750. Try a discount rate of 19 percent. The NPV at 19 percent discount rate is:
– Rs 780.

Hence the IRR is:

$$18\% + \frac{1750}{2530} \times 1\% \\ = 18.69\%$$

MIRR AND PAYBACK

d. The future value of benefits when compounded at 12 percent is:

$$20,000 (1.12)^4 + 30,000 (1.12)^3 + 40,000 (1.12)^2 + 50,000 (1.12) + 30,000 \\ = 209,790$$

$$100,000 (1 + r^*)^5 = 209,790$$

$$r^* = 15.97\%$$

Hence the MIRR is 15.97%

e. The payback period is slightly more than 3 years.

f. The discounted payback period is slightly less than 4 years.

PROBLEM 1

Your company is considering two mutually exclusive projects, A and B. Project A involves an outlay of ₹100 million which will generate an expected cash inflow of ₹25 million per year for 6 years. Project B calls for an outlay of ₹50 million which will produce an expected cash inflow of ₹13 million per year for 6 years. The company's cost of capital is 12 percent.

- (a) Calculate the NPV and IRR of each project.
- (b) What is the NPV and IRR of the differential project (the project that reflects the difference between Project B and Project A).

PROBLEM 2

Your company is considering two projects, Project M and Project N, each of which requires an initial outlay of ₹50 million. The expected cash inflows from these projects are:

<i>Year</i>	<i>Project M</i>	<i>Project N</i>
1	₹11 million	₹38 million
2	19	22
3	32	18
4	37	10

- (a) What is the payback period for each of the projects?
- (b) What is the discounted payback period for each of the projects if the cost of capital is 12 percent?
- (c) If the two projects are independent and the cost of capital is 12 percent, which project(s) should the firm invest in?
- (d) If the two projects are mutually exclusive and the cost of capital is 10 percent, which project should the firm invest in?
- (e) If the two projects are mutually exclusive and the cost of capital is 15 percent, which project should the firm invest in?
- (f) If the cost of capital is 14 percent, what is the modified IRR of each project?